Most important MCQS ACCORDING TO CURRENT PAPER 2025

Important MCQs from Lectures 1–18 (CS504 – Software Engineering I)

1. Software Engineering is concerned with _____
   - A) Computers only
   - B) Programming only
   - C) Theoretical aspects only
   - D) All aspects of software production
2. Software process is a set of activities that leads to the production of _____.
   - A) Hardware
   - B) Software product
   - C) Requirements
   - D) Design
3. Which of the following is not a software process model?
   - A) Waterfall
   - B) Incremental
   - C) Spiral staircase
   - D) Spiral
4. The main drawback of the waterfall model is:
   - A) Too simple
   - B) Inflexibility in accommodating changes
   - C) No testing
   - D) No planning
5. A software process includes:
   - A) Specification
   - B) Development
   - C) Validation
   - D) All of the above

6. The Spiral model was proposed by:
   - A) Pressman
   - B) Boehm
   - C) Sommerville
   - D) Royce
7. The agile model emphasizes:
   - A) Heavy documentation
   - B) Rapid delivery of working software
   - C) Formal methods
   - D) Big design upfront
8. Which process model is best suited for small projects with clear objectives?
   - A) Waterfall
   - B) Spiral

Learning with Amie

- C) RAD
- D) Agile

9. In which model is risk management a prominent feature?
    - A) Agile
    - B) Waterfall
    - C) Spiral
    - D) Prototype
10. RAD model stands for:

- A) Rapid Automated Development
- B) Rapid Application Development
- C) Relative Application Development
- D) Real-time App Development

11. Project scheduling involves:

- A) Estimating the time required
- B) Coding
- C) Marketing
- D) Maintenance

12. Which chart is used in project scheduling?

- A) Pie chart
- B) Bar chart
- C) Gantt chart
- D) Line chart

13. The primary purpose of project planning is:

- A) To manage resources and schedule
- B) To design code
- C) To test
- D) To maintain

14. A project plan includes:

- A) Schedule
- B) Risk management
- C) Resources
- D) All of the above

15. Risk management helps in:

- A) Identifying and mitigating future problems
- B) Writing code faster

Learning with Annie

- C) Increasing revenue
- D) Designing UI

16. A requirement is a:

- A) Constraint
- B) Statement of what the system should do
- C) User interface
- D) Test case

17. Functional requirements describe:

- A) Performance
- B) What the system should do
- C) Look and feel
- D) Hardware

18. Non-functional requirements include:

- A) Inputs
- B) Performance, usability
- C) Outputs
- D) Tables

19. SRS stands for:

- A) Software Ready System
- B) Software Requirements Specification
- C) System Requirements Schedule
- D) Software Request Sheet

20. The process of finding out, analyzing, documenting, and checking services is called:

- A) Planning
- B) Requirements engineering
- C) Coding
- D) Designing

21. Technical feasibility assesses:

- A) Financial benefits
- B) Technical resources & skills
- C) Legal issues
- D) Marketing strategies

22. The three main types of feasibility are:

Learning with Annie

- A) Financial, Logical, Practical
- B) Technical, Economic, Operational
- C) Logical, Technical, Functional
- D) Operational, Functional, Commercial

23. DFD stands for:

- A) Data Flow Diagram
- B) Design Flow Document
- C) Data Function Design
- D) Document Flow Diagram

24. In DFDs, circles represent:

- A) Data store
- B) Process
- C) Data flow
- D) Entity

25. An external entity in a DFD is represented by:

- A) Rectangle
- B) Circle
- C) Arrow
- D) Diamond

26. Context Diagram shows:

- A) The system as a single process
- B) Detailed DFDs
- C) UML classes
- D) Use cases

27. ERD stands for:

- A) Entity Requirement Diagram
- B) Entity Relationship Diagram
- C) Entity Role Diagram
- D) Enhanced Relationship Design

28. UML stands for:

- A) Unified Modeling Language
- B) Unified Machine Language
- C) Unique Model Layout
- D) User Mode Layout

Learning with Annie

29. In UML, a class is represented by:

- A) Circle
- B) Rectangle
- C) Arrow
- D) Triangle

30. Use case diagram shows:

- A) Interaction between actors and system
- B) Class hierarchy
- C) State machines
- D) Flowcharts

31. Which model is best for evolving requirements?

- A) Spiral
- B) Waterfall
- C) RAD
- D) V-Model

32. Validation ensures:

- A) The system runs fast
- B) The system meets the requirements
- C) Code is optimized
- D) UI is beautiful

33. Verification ensures:

- A) The product is built correctly
- B) System is ready for market
- C) Performance is good
- D) Data is backed up

34. In Gantt charts, time is shown on the:

- A) Vertical axis
- B) Horizontal axis
- C) Diagonal line
- D) Pie slices

35. The main focus of software engineering is:

- A) Hardware efficiency
- B) Software quality and productivity

Learning with Annie

- C) Marketing
- D) Design only

36. Agile promotes:

- A) Customer collaboration
- B) Heavy documentation
- C) Long-term planning only
- D) Code-first approaches

37. Software crisis refers to:

- A) Inability to deliver quality software on time
- B) Running out of RAM
- C) Lack of hardware
- D) Internet speed

38. Requirements should be:

- A) Vague
- B) Complete and unambiguous
- C) Estimated
- D) Short

39. A well-known tool for project estimation:

- A) Function Points
- B) UML
- C) DFD
- D) Use Cases

40. IEEE stands for:

- A) Institute of Electrical and Electronics Engineers
- B) Institution of Electronic Engineers
- C) Industrial Electrical Engineering
- D) International Engineers' Entity

41. High-level design is also called:

- A) Architectural design
- B) Coding
- C) UI design
- D) Object design

42. A prototype is built to:

Learning with Annie

- A) Understand customer requirements
- B) Speed up coding
- C) Avoid testing
- D) Save costs

43. The final product is delivered in:

- A) Waterfall phase 1
- B) Last phase
- C) Spiral iteration
- D) Initial prototype

44. Most abstract DFD is:

- A) Level 0
- B) Level 1
- C) Context level
- D) Logical DFD

45. Detailed design focuses on:

- A) UI only
- B) Internal working of modules
- C) Documentation
- D) Planning

46. SQA stands for:

- A) Software Quality Assurance
- B) System Quality Analysis
- C) Software Queue Assessment
- D) Software Quick Access

47. Software reuse is promoted in:

- A) Waterfall
- B) Component-based model
- C) Agile only
- D) Spiral only

48. A milestone is:

- A) A significant point in project timeline
- B) A task
- C) A defect
- D) A delay

Learning with Annie

49. Coding phase transforms:

- A) Hardware to software
- B) Design into executable programs
- C) Documents to design
- D) Plan to test

50. UML diagrams include:

- A) Use case
- B) Class
- C) Sequence
- D) All of the above

"علم ایک نور ہے جو جہالت کے اندھیرے کو ہدور کرتا ہے"

learning with Anie